

Software Requirements Specification

Campus Event Management System

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Date: [15/03/2026](#)

1. Introduction

The **Campus Event Management System (CEMS)** is a web-based platform designed to help universities and colleges plan, organize, manage, and promote campus events efficiently. The system enables students, faculty members, and administrators to create events, register for events, receive notifications, and manage event details in a centralized digital platform.

The system reduces manual work involved in organizing campus events such as seminars, workshops, cultural festivals, and club activities. By providing automated notifications, event registration management, and event tracking features, the platform improves communication and coordination within the campus community.

1.1 Document Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements of the **Campus Event Management System**. It provides a detailed overview of the system's features, constraints, design considerations, and operational requirements.

The purpose of this document is to serve as a reference for developers, testers, project managers, and stakeholders during the development and implementation of the system. It ensures that all team members clearly understand the system requirements before development begins.

1.2 Product Scope

The **Campus Event Management System** is a web application designed to manage and streamline the process of organizing events in a university or college environment. It allows event organizers to create events, manage event details, track participants, and communicate with attendees.

The system also allows students to browse upcoming events, register for events, receive notifications about event updates, and track their participation history. Administrators have control over approving events, monitoring event activities, and managing users.

Benefits of the system

- Centralized event management platform
- Easy event creation and registration
- Automatic notifications and reminders
- Better coordination between organizers and students
- Improved visibility of campus activities

1.3 Intended Audience and Document Overview

This document is intended for the following audiences:

Developers

Developers will use this document to understand system requirements and implement system functionality.

Project Managers

Project managers will use this document to monitor development progress and ensure that requirements are met.

Testers

Testers will use this document to create test cases and verify system functionality.

Clients / Professors

Clients or instructors will use this document to verify that the system meets the expected project requirements.

Document Overview

The document is organized as follows:

Section 1 – Introduction and overview of the system

Section 2 – Overall system description

Section 3 – Detailed functional requirements

Section 4 – Non-functional requirements

Section 5 – Additional requirements

Appendices – Data dictionary and group activity logs

1.4 Definitions, Acronyms and Abbreviations

Term	Meaning
CEMS	Campus Event Management System
UI	User Interface
DB	Database
API	Application Programming Interface
Admin	System Administrator
User	Student or faculty using the system
CRUD	Create, Read, Update, Delete operations
React	JavaScript library for building user interfaces
Node.js	Backend runtime environment
PostgreSQL	Relational database management system

1.5 Document Conventions

The document follows **IEEE Software Requirements Specification standards**.

Formatting conventions used:

- Font: Arial
- Font Size: 11 or 12
- Headings organized using numbered sections

- Functional requirements numbered as **F1, F2, F3...**
- Use cases numbered as **U1, U2, U3...**

1.6 References and Acknowledgments

References used for preparing this document include:

- IEEE Software Requirements Specification Guidelines
- React Documentation – <https://react.dev>
- Node.js Documentation – <https://nodejs.org>
- PostgreSQL Documentation – <https://postgresql.org>

2 Overall Description

2.1 Product Overview

The **Campus Event Management System** is a web-based application designed to support campus event planning and management. It provides an interface for event organizers to create and manage events and for students to discover and participate in those events.

The system uses the following technology stack:

Frontend: React

Backend: Node.js

Database: PostgreSQL

System Components

User Interface (React)

Backend Server (Node.js)

Database (PostgreSQL)

Notification Service

System Interaction Diagram (conceptual)

Users (Students / Organizers / Admin)

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React Web Application

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Node.js Backend Server

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PostgreSQL Database

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Notification System (Email / Alerts)

2.2 Product Functionality

Major functions of the system include:

- User registration and login
- Event creation and management
- Event browsing and searching
- Event registration for students
- Notification system for upcoming events
- Event approval by administrators
- Dashboard for event statistics
- Event reminders and updates
- Event cancellation and modification

2.3 Design and Implementation Constraints

The following constraints apply to the system development:

- The frontend must be implemented using **React**
- The backend must be implemented using **Node.js**
- The database must use **PostgreSQL**
- The system must support modern web browsers
- UML diagrams must be used for system modeling

- REST API architecture must be followed
- The system must ensure secure user authentication

2.4 Assumptions and Dependencies

Assumptions:

- Users have internet access
- Students have valid university email accounts
- Administrators manage event approvals
- Users have basic knowledge of web applications

Dependencies:

- React framework
- Node.js runtime environment
- PostgreSQL database server
- Email notification services

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system will provide a **web-based graphical user interface** accessible through browsers.

Main UI Components:

- Login and Registration Page
- Event Dashboard
- Event Creation Form
- Event Details Page

- Event Registration Page
- Admin Management Panel

Users will interact with the system through forms, buttons, navigation menus, and dashboards.

3.1.2 Hardware Interfaces

The system interacts with:

- User devices (Laptop, Desktop, Mobile browser)
- Web servers
- Database servers

No specialized hardware is required.

3.1.3 Software Interfaces

The system interacts with:

- PostgreSQL Database
- Email Notification Services
- Authentication libraries
- Web browser clients

3.2 Functional Requirements

F1 User Registration

The system shall allow users to create an account using email and password.

F2 User Login

The system shall allow registered users to log into the system.

F3 Event Creation

The system shall allow event organizers to create events.

F4 Event Editing

Organizers shall be able to edit event details.

F5 Event Deletion

Organizers shall be able to delete events.

F6 Event Search

Users shall be able to search events by category, date, or keyword.

F7 Event Registration

Users shall be able to register for events.

F8 Notifications

The system shall notify users about upcoming events.

F9 Event Approval

Administrators shall approve or reject events.

F10 Event Dashboard

Users shall view upcoming and registered events.

3.3 Use Case Model

Actors

- Student
- Event Organizer
- Administrator

Use Case U1 – Register for Event

Author: Team Member

Purpose: Allows students to register for an event.

Priority: High

Actors: Student

Preconditions:

- Student must be logged in
- Event must be available

Postconditions:

- Student is successfully registered

Basic Flow:

- 1 Student logs into system
- 2 Student selects event
- 3 Student clicks register
- 4 System stores registration
- 5 Confirmation message displayed

Alternative Flow:

Event full → Registration denied.

Exceptions:

Network error → Registration fails.

Use Case U2 – Create Event

Actor: Event Organizer

Basic Flow:

- 1 Organizer logs in
- 2 Selects "Create Event"
- 3 Enters event details
- 4 Submits form
- 5 Event saved in database

4 Other Non-functional Requirements

4.1 Performance Requirements

P1 The system should support at least **500 concurrent users**.

P2 Event pages must load within **3 seconds**.

P3 Event notifications must be delivered within **1 minute** of triggering.

4.2 Safety and Security Requirements

- Users must authenticate using secure login credentials
- Passwords must be encrypted
- Access control must restrict admin functions

- Database must prevent unauthorized access
- HTTPS protocol must be used

4.3 Software Quality Attributes

Reliability

The system must maintain consistent operation with minimal downtime.

Usability

The interface must be easy to navigate for students and administrators.

Maintainability

Code should be modular to allow easy updates and bug fixes.

Scalability

The system should support increasing numbers of events and users.

5 Other Requirements

Database Requirements:

Main database tables include:

- Users
- Events
- Event Registrations
- Notifications

The system must maintain proper relational database design.

Appendix A – Data Dictionary

Data Name	Description
UserID	Unique identifier for users
EventID	Unique identifier for events
EventName	Name of the event
EventDate	Date of the event
Location	Event venue
OrganizerID	User ID of event organizer
RegistrationID	Unique ID for event registration

Appendix B – Group Log

Example entries:

Meeting 1 – Project topic selection

Meeting 2 – Requirement analysis

Meeting 3 – SRS document preparation